

SHI FENG

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EDUCATION

Tsinghua University

Aug 2019 - Jul 2023 (expected)

Bachelor of Science, selected to Yao Class

- GPA: 3.87/4.0
- GRE Scores: Verbal Reasoning 157, Quantitative Reasoning 170, Analytical Writing 4.0
- TOEFL Best Scores: Reading 30, Listening 29, Speaking 22, Writing 29
- Selected Courses: Introduction to AI (A), Mathematics for Computer Science (A), Causal and Statistical Learning (A+), Abstract Algebra (A), Introduction to Databases (A+), Artificial Intelligence: Principles and Techniques (A), Theory of Computation (A), Distributed and Blockchain System (A-), Machine Learning (A), Operating System (A), Research Immersion Training (A+)

RESEARCH INTERESTS

Theoretical computer science and economics, particularly algorithmic game theory, learning theory, causality, and machine learning.

RESEARCH EXPERIENCES

Multi-Agent Owner-Assisted Scoring Mechanisms

Aug 2022 - present

Advisors: Prof. [Yuhao Wang](#), Prof. [Weijie Su](#)

Tsinghua IIIS&UPenn Wharton School

- Extended owner-assisted scoring problem to a multi-agent setting and wrote a research proposal for it.
- Designed a fair, “exchangeable noise assumption free”, and truthful mechanism for the extended problem.

Peer Prediction for Learning Agents

Feb 2022 - Jun 2022

Advisors: Prof. [Yiling Chen](#), Prof. [Fang-Yi Yu](#)

Harvard EconCS Group (in-person)

- Surveyed mechanisms for information elicitation and extended the original multi-task setting to a sequential setting.
- Proved truthful convergence of correlated agreement mechanism adopted on agents using reward-based online learning algorithms and a negative result showing no-regret assumption on agents can-not guarantee truthful convergence.
- Created a peer prediction simulator and conducted numerical experiments which supported theoretical results.

Combinatorial Causal Bandits

Jul 2021 - Feb 2022

Advisor: Prof. [Wei Chen](#)

MSR Asia Theory Center

- Surveyed causal bandits algorithms for pure exploration and cumulative regret minimization.
- Presented the new combinatorial causal bandit framework with a definition of cumulative regret.
- Proposed algorithms for combinatorial causal bandits problem on binary generalized linear models with hidden variables.
- Implemented algorithms in OOP and conducted numerical experiments to show their efficiency.

Causal Inference for Social Networks

Nov 2020 - Jul 2021

Advisor: Prof. Wei Chen

MSR Asia Theory Center

- Surveyed identifiability of Bayesian causal models and completeness of the do-calculus algorithm.
- Provided transformations from an Independent Cascade Model to a Bayesian causal model.
- Proved necessary and sufficient conditions for identifiability in three common types of Independent Cascade Models.

PUBLICATIONS

(* denotes equal contribution)

Shi Feng, Wei Chen. Combinatorial Causal Bandits.

The 37th AAAI Conference on Artificial Intelligence (AAAI 2023).

[\[arXiv\]](#) [\[code\]](#)

Shi Feng, Fang-Yi Yu, Yiling Chen. Peer Prediction for Learning Agents.

The 36th Conference on Neural Information Processing Systems (NeurIPS 2022).

[\[arXiv\]](#) [\[code\]](#)

Shi Feng, Wei Chen. Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model.

The 10th International Conference on Computational Data and Social Networks (CSoNet 2021) **Best Paper**.

[\[arXiv\]](#) [\[publication\]](#)

Shi Feng*, Zimeng Song*, Weijie Su, Yuhao Wang. A Truthful Multi-Agent Owner-Assisted Scoring Mechanism.

Writing in progress.

[\[slides\]](#)

SELECTED HONORS & AWARDS

Comprehensive Excellence Awards (top 10%)

Oct 2020, Oct 2022, THU

“Star of Tomorrow” Award of Excellence

Mar 2022, MSR Asia

Best Paper Award (1/57)	<i>Nov 2021, CSoNet</i>
Innovation Excellence Award	<i>Oct 2021, THU</i>
Social Practice Excellence Award	<i>Oct 2021, THU</i>
“Challenge Cup” Science and Technology Competition, Top Grade Prize (top 3%)	<i>Apr 2021, THU</i>
Freshman Scholarship (top 10%)	<i>Oct 2019, THU</i>
Tsinghua Xuetaang Scholarship	<i>Aug 2019, THU</i>
Russian Mathematical Olympiad, Silver Medal	<i>Apr 2019, MSE of Russia</i>
Chinese Mathematical Olympiad, Gold Medal	<i>Nov 2018, CMS</i>
National High School Mathematics League, First Prizes	<i>Sept 2016, Sept 2017, Sept 2018, CMS</i>

TALKS

NeurIPS 2022 , Peer Prediction for Learning Agents	<i>Oct 2022</i>
Yao Class Seminar , Peer Prediction for Learning Agents	<i>Sept 2022</i>
CSoNet 2021 , Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model	<i>Nov 2021</i>

SERVICES

Class Leader , Yao Class 91, Tsinghua University	<i>Aug 2022 - present</i>
Reviewer , Journal of Combinatorial Optimization (JOCO)	<i>May 2022</i>
Team Member , Chinese national team for Russian Mathematical Olympiad	<i>Apr 2019</i>

SKILLS

Programming Languages: C/C++, Python, Golang, SQL, assembly language, Bash, HTML

Professional Applications: Latex, Mathematica, Matlab, PyTorch, TensorFlow, Git

Languages: Chinese (native language), English (fluent)

Hobbies: soccer (member of IIS soccer team), road trip (driving through China National Highway 318 in 2019), handball (amateur training for one year), basketball, vehicle engineering (1 invention patent and 1 utility model patent), Go (amateur 1 dan)